American University of Beirut
EECE 502 – Final Year Project
Course Coordinator – Dr. Youssef Tawk
Minutes for the Weekly Group Meeting – Submitted per Group

Meeting #	Date:	Time:	Duration:	Meeting called by:	Minutes Taker:
12	1/4/26	9 am	50min	◉ Advisor ○ Students	Hadi Elham

Attendees:

Raghad Al Agha, Hadi Elham, Lynn Ariss, Claritta Khoury, Prof. Ayman Kayssi

Briefly summarize the main discussions during the meeting:

The meeting focused on continued development of the mobile application, machine-learning modeling, and hardware system optimization. On the app side, the team worked on refining the UI and ensuring that all previously implemented features remain functional before adding new ones, including updates such as language support and onboarding improvements. A new interactive tutorial system was designed to appear as a scrollable pop-up instead of a static document, allowing users to navigate or skip steps easily, and it is planned to be triggered after sign-up. On the modeling side, the team analyzed inconsistencies in prior research papers, particularly regarding dataset splitting and augmentation, and explored alternative approaches using pretrained models such as VGG19 combined with classifiers (SVM, KNN, ANN). They achieved improved accuracy (~86%) compared to previous models and continued refining training strategies using proper dataset splits (68/17/15). Additionally, work was done on scaling down the hardware system by reducing wiring complexity and reorganizing multiplexers to make the design more compact and wearable. The team tested different configurations for placing multiplexers on the foot and discussed removing resistors to simplify connections, though this raised concerns about circuit correctness.

Briefly summarize the conclusions drawn regarding the above-mentioned discussions:

The team concluded that the application is stable at its current stage and should be verified before introducing additional features, with the interactive tutorial expected to improve usability. The machine-learning pipeline is improving through more reliable methodologies and exploration of alternative architectures, though further validation is still needed. On the hardware side, system miniaturization is progressing, but electrical constraints—such as the necessity of resistors and proper multiplexer behavior—must be carefully respected based on datasheet specifications. The next steps include validating comfort and feasibility of the compact hardware layout, finalizing circuit integrity, and continuing improvements in both modeling accuracy and system integration.

Enumerate the assigned tasks by the FYP advisor for each student + the deadline for delivery:

Assigned Task / Per Student	Name of the Student	Deadline
Work to enhance ataxia model, impleemnt charging circuit, integrate ppg detection algorithm and IMU use in project	Hadi Elham	8/4/2026
Continue wokring on UI, work on app tutorial	Lynn Ariss	8/4/2026
Continue wokring on UI, work on app tutorial	Raghad Al Agha	8/4/2026
Scaling down the system, work on datasheet	Claritta Khoury	8/4/2026